

Deep Learning for Solving Dynamic Models

Session 3a: Neural architecture search, random search and Hyperband

Simon Scheidegger

HEC, University of Lausanne

University of Padova

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Roadmap of This Lecture

I. Motivation

- ▶ Why architecture matters
- ▶ The hyperparameter space
- ▶ Limits of manual tuning

II. Search Methods

- ▶ Grid vs. random search
- ▶ Bayesian optimization
- ▶ Hyperband / successive halving
- ▶ Method comparison

III. Implementing the Search

- ▶ Defining a search space
- ▶ Random Search + SHA from scratch
- ▶ Tool landscape (footnote)

IV. Example & Wrap-up

- ▶ Genz-function approximation
- ▶ Best practices
- ▶ Connection to DEQNs

Hands-on: 02_NAS_Random_Search_10D.ipynb (10-D random search)

03_NAS_RandomSearch_Hyperband.ipynb (Random Search + SHA)

Learning objectives

By the end of this lecture you should be able to:

- ▶ Diagnose architecture sensitivity and quantify how network depth and width affect approximation error
- ▶ Compare grid search, random search, Bayesian optimization, and Hyperband on their computational cost and sample efficiency
- ▶ Implement random search and successive halving from scratch in plain Python
- ▶ Apply NAS to function approximation problems and choose appropriate search spaces for economic models
- ▶ Select production tools (KerasTuner, Optuna, Ray Tune) and apply best practices to scale architecture search efficiently

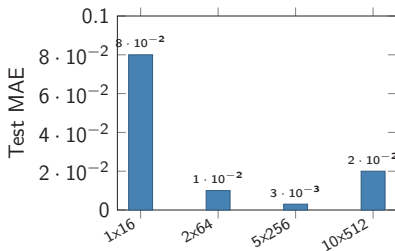
Why Architecture Matters

Same data, different architectures:

- ▶ 1 layer, 16 units $\Rightarrow$ MAE = 0.08
- ▶ 2 layers, 64 units $\Rightarrow$ MAE = 0.01
- ▶ 5 layers, 256 units $\Rightarrow$ MAE = 0.003
- ▶ 10 layers, 512 units $\Rightarrow$ MAE = 0.02 (overfit)

Architecture choice can matter **more than training duration**.

Illustrative numbers from a depth/width sweep on the companion regression task; the from-scratch search in Notebook 03 reproduces the same ordering on a different testbed.



The Hyperparameter Space

Architectural hyperparameters:

- ▶ Number of hidden layers
 $L \in \{1, \dots, 5\}$
- ▶ Units per layer $n_\ell \in \{32, 64, \dots, 256\}$
(8 values)
- ▶ Activation: ReLU, tanh, swish

Training hyperparameters:

- ▶ Learning rate η log-uniform on
 $[10^{-4}, 10^{-2}]$ (20-point grid)
- ▶ Batch size $B \in \{16, 32, 64, 128\}$
- ▶ Epochs, regularization, ...

Combinatorial explosion

Even this modest search space:

$$|\mathcal{H}| = 5 \times 8 \times 3 \times 20 \times 4 = 9,600$$

- ▶ Each trial: minutes to hours
- ▶ Full grid: **infeasible**

Manual Tuning

The “grad-student descent” approach:

1. Pick an architecture (intuition / paper)
2. Train, evaluate, adjust one thing
3. Repeat until deadline

Problems:

- ▶ **Doesn't scale** beyond a few hyperparameters
- ▶ **Human bias**: we try what we already know
- ▶ **Not reproducible**: results depend on the researcher
- ▶ Misses interactions between hyperparameters

Key insight

We need **automated, systematic** methods to search the hyperparameter space efficiently.

Automated Search Strategies

Grid Search

Idea: Enumerate all combinations on a regular grid.

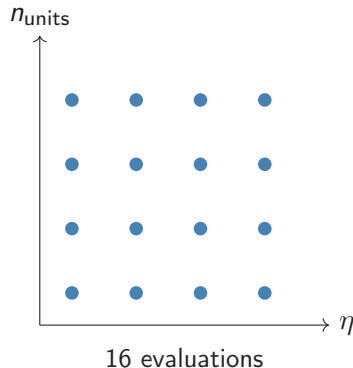
Algorithm

1. Define discrete set for each HP
2. Evaluate *every* combination
3. Return the best

Cost: $\mathcal{O}(k^d)$ — exponential in dimension d .

Pros: Simple, embarrassingly parallel.

Cons: Wastes budget on **unimportant** dimensions.



Random Search

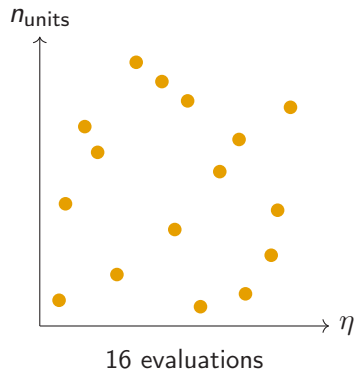
Idea: Sample configurations uniformly at random.

Key result (Bergstra & Bengio, 2012)

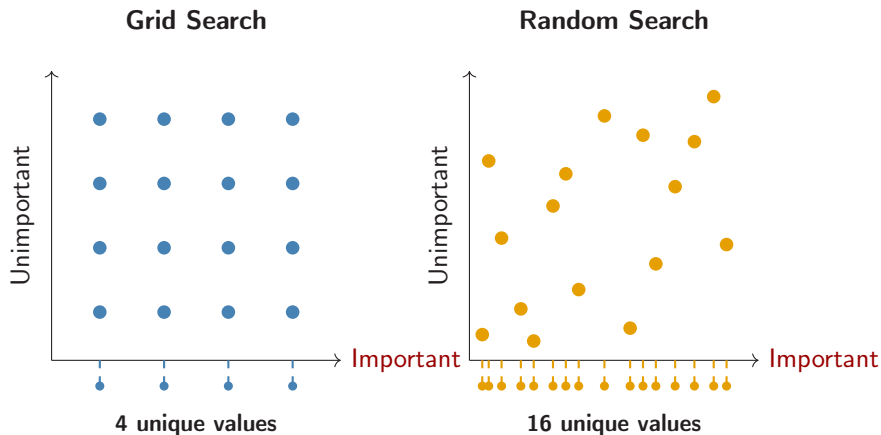
When only a *few* dimensions matter, random search finds good configurations with **fewer trials** than grid search.

Why? Random points explore each dimension *independently* — more unique values per HP.

Cost: $\mathcal{O}(n)$ — linear in budget n .



Grid vs. Random — Why Random Wins



When one dimension matters, the grid wastes most evaluations on the unimportant axis; random search samples the important axis more densely.

Bayesian Optimization

Idea: Build a *surrogate model* of $f(\text{HP}) \rightarrow \text{loss}$ and use it to choose the next trial *intelligently*.

1. Fit a Gaussian process (GP) to observed trials
2. Compute an **acquisition function** (e.g., Expected Improvement)
3. Evaluate the HP with highest acquisition value
4. Update the GP and repeat

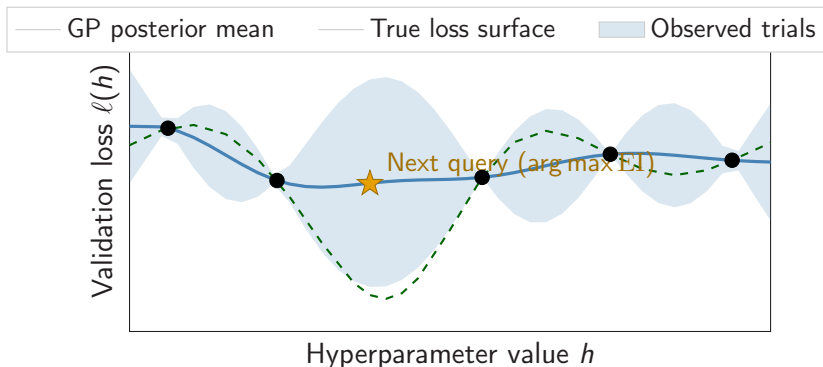
Key advantage: **Sample-efficient** — exploits structure in the response surface.

Expected Improvement

$$\text{EI}(\mathbf{x}) = \mathbb{E}[\max(f^* - f(\mathbf{x}), 0)]$$

- ▶ f^* : best observed value
- ▶ Balances *exploration* (high uncertainty) and *exploitation* (low predicted loss)

Bayesian Optimization — Visual



The GP posterior mean *interpolates* the five observations and is *wide* in unexplored gaps. The Expected-Improvement acquisition function peaks at $h \approx 3.75$, trading off the high posterior variance there against the moderately low predicted mean.

Hyperband

Idea: Start many configurations cheaply; **stop the worst early** and give more resources to promising ones.

Successive Halving (SHA, $\eta = 3$)

1. Start n random configs with budget b
2. Train each for b epochs, evaluate
3. **Keep the top $1/\eta$** of configs (discard the rest)
4. Multiply the budget by η , repeat

Hyperband = SHA with several brackets (different n/b trade-offs), each bracket itself being a stage cascade like $9 \rightarrow 3 \rightarrow 1$ above. The usual default is $\eta = 3$.

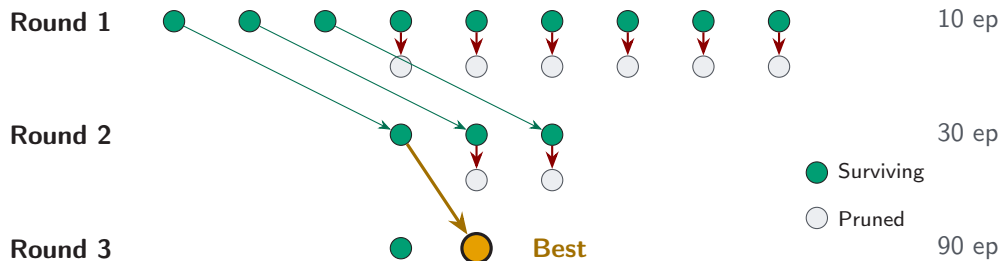
Resource use: many cheap trials; full budgets only for survivors.

Key insight:

- ▶ Bad configs identifiable *early*
- ▶ No need for 200 epochs to know $\eta=10$ diverges
- ▶ Exploits **correlation** between early and late performance

Li et al. (2018): large speedups when early and late performance are correlated.

Hyperband — Successive Halving



$9 \rightarrow 3 \rightarrow 1$ (factor $\eta = 3$): total cost = $9 \times 10 + 3 \times 30 + 1 \times 90 = 270$ epoch-equivalents (vs. $9 \times 90 = 810$ for full training).

Bracket vs. schedule. This $9 \rightarrow 3 \rightarrow 1$ cascade is one SHA bracket. Hyperband itself runs several brackets in parallel, each with a different starting (n, b) trade-off, hedging across the unknown correlation between early- and late-training performance.

Method Comparison

Property	Grid	Random	Bayesian	Hyperband
Computational cost	$\mathcal{O}(k^d)$	$\mathcal{O}(n)$	$\mathcal{O}(n)$	$\mathcal{O}(n \log n)$
Sample efficiency	Low	Medium	High	High
Parallelisable	Fully	Fully	Limited	Partially
Handles interactions	No	Partially	Yes	No
Early stopping	No	No	No	Yes
Implementation	Trivial	Trivial	Moderate	Moderate

Practical advice:

- ▶ **Quick exploration:** Random search (strong baseline)
- ▶ **Limited budget:** Bayesian optimization (sample-efficient)
- ▶ **Large search space:** Hyperband (aggressive early stopping)

Implementing the Search

Random Search and Successive Halving, from scratch

Defining a Search Space

A plain dict — no library required

```
SPACE = {
    "layers":      [1, 2, 3, 4, 5],
    "units":       [32, 64, 96, 128, 160, 192, 224, 256],
    "activation":  ["relu", "tanh", "swish"],
    "lr":          ("log_uniform", 1e-4, 1e-2),
}

def sample_config(rng):
    return {
        "layers":      int(rng.choice(SPACE["layers"])),
        "units":       int(rng.choice(SPACE["units"])),
        "activation":  str(rng.choice(SPACE["activation"])),
        "lr":          float(10 ** rng.uniform(-4, -2)),
    }
```

No `hp.Int` / `hp.Float` abstraction — the search space is a dict, the sampler is one function.

Random Search from Scratch

30 trials, single function

```
def random_search(n_trials=30, epochs=50, seed=0):
    rng = np.random.default_rng(seed)
    records = []
    for t in range(n_trials):
        cfg = sample_config(rng)
        model = build_model(cfg)
        h = model.fit(X_tr, y_tr, epochs=epochs,
                      validation_data=(X_val, y_val), verbose=0)
        records.append((cfg, min(h.history["val_loss"])))
    return min(records, key=lambda r: r[1])
```

Independent uniform draws; $\mathcal{O}(N)$ cost, embarrassingly parallel (Bergstra & Bengio, 2012).

Successive Halving from Scratch

$n_0 = 27$, $\eta = 3$, three rounds

```
def successive_halving(n0=27, eta=3, r0=8, seed=0):
    rng = np.random.default_rng(seed)
    survivors = [(sample_config(rng), None) for _ in range(n0)]
    r = r0
    while len(survivors) > 1:
        scored = [(cfg, train_for(cfg, epochs=r)) for cfg, _ in survivors]
        scored.sort(key=lambda s: s[1])                # ascending val loss
        keep    = max(1, len(scored) // eta)
        survivors = scored[:keep]
        r *= eta                                       # 8 -> 24 -> 72 epochs
    return survivors[0]
```

Notebook schedule: 27 candidates @ 8 ep $\rightarrow$ 9 @ 24 ep $\rightarrow$ 3 @ 72 ep $\rightarrow$ 1 winner. This is 648 config-epochs, versus 1500 for 30 random-search trials at 50 epochs each: about $2.3\times$ less compute while testing a comparable number of architectures.

Production Tooling (footnote)

We just wrote both algorithms in ~ 25 lines of Python. Real projects rarely hand-roll the loop; established libraries wrap (and parallelise) the same algorithms behind a uniform API:

Library	Algorithms	Notes
KerasTuner	Random, Bayesian, Hyperband	Tight Keras integration
Optuna	TPE, CMA-ES, Hyperband, NSGA-II	Framework-agnostic; popular default
Ray Tune	All of the above + ASHA, PBT	Distributed by design
Hyperopt	TPE, Random	The original TPE reference
Ax / BoTorch	Bayesian (multi-objective, MOO)	Meta; PyTorch-native GPs
NNI	Random, Bayesian, evolutionary, NAS	Microsoft; full graph-NAS support

Why teach algorithms, not wrappers. APIs change; the search loops (Random, SHA/Hyperband, GP+EI, TPE) persist.

Example: Genz Function

Approximation

Experiment Setup

Target function: Genz Gaussian on $[0, 1]^2$

$$f(\mathbf{x}) = \exp\left(-\sum_{i=1}^d c_i^2 (x_i - w_i)^2\right)$$

Data:

- ▶ $d = 2$, uniform random samples
- ▶ $n_{\text{train}} = 1,000$, $n_{\text{test}} = 2,000$
- ▶ Loss: MSE, metric: test MAE

Search space:

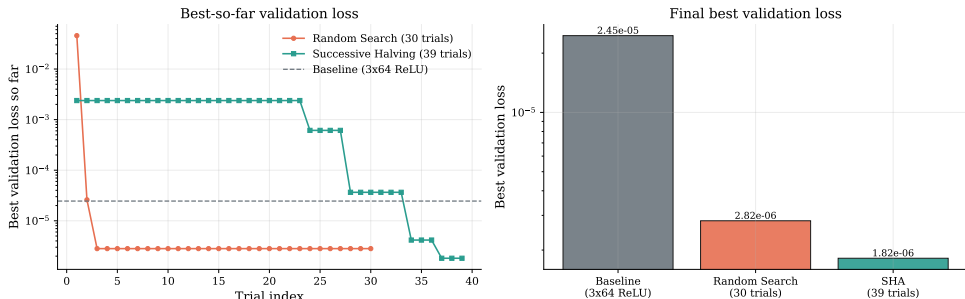
- ▶ Layers: 1–5
- ▶ Units/layer: 32–256 (step 32)
- ▶ Activation: ReLU, tanh, swish
- ▶ LR: 10^{-4} – 10^{-2} (log scale)

Baseline: 3×64 ReLU MLP, $\eta = 10^{-3}$

Budget: 30 trials per method

Search Results (Random Search vs. SHA on a 2-D Genz Gaussian)

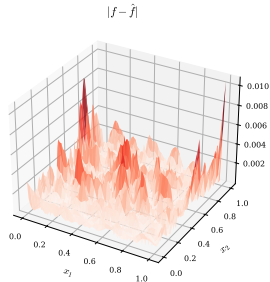
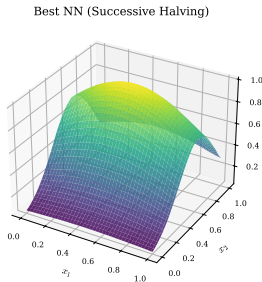
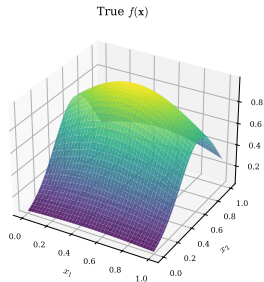
NAS on Genz Gaussian ($d=2$): Random Search vs Successive Halving



Generated by 03_NAS_RandomSearch_Hyperband.ipynb. Random Search uses 30 full trials; SHA uses 27 initial candidates with $\eta = 3$. Both reach MAE of order 10^{-3} from a baseline near $4 \cdot 10^{-3}$; SHA uses about $2.3\times$ less config-epoch budget in this seeded run.

Best Model vs. Ground Truth

Best NAS model vs ground truth (MAE = 0.00099)



Generated by `03_NAS_RandomSearch_Hyperband.ipynb`. Left to right: the true Genz target, the best NAS-found NN approximation, and the absolute error map on $[0, 1]^2$. No placeholder shapes; this is the actual notebook output. The companion `02_NAS_Random_Search_10D.ipynb` offers a 10-D testbed (search loop in plain Python, model in TF/Keras).

Best Practices for NAS

Search space design:

- ▶ Start small, expand if needed
- ▶ Use **log-scale** for learning rate
- ▶ Constrain to reasonable ranges
- ▶ Include the current best as a point

Budget allocation:

- ▶ Diminishing returns after ~ 30 trials
- ▶ Hyperband: set factor=3 (default)
- ▶ Use early stopping callbacks

Common pitfalls:

- ▶ Search space too large $\Rightarrow$ wasted budget
- ▶ Overfitting to validation set $\Rightarrow$ use holdout test
- ▶ Ignoring training stability $\Rightarrow$ check learning curves

Rule of thumb

If manual tuning took N trials, automated search with $3N$ trials will likely find something better.

Connection to Economics

Where NAS matters in computational economics:

1. **Surrogate models:** Replace expensive simulations (e.g., climate IAMs) with NNs — architecture determines approximation quality.
2. **DSGE policy functions:** Deep equilibrium nets (Lecture 03) use NNs to approximate policy/value functions. **Architecture search over the policy network** can improve convergence.
3. **High-dimensional problems:** As d grows (many state variables), the architecture must scale. NAS automates this adaptation.
4. **Reproducibility:** Automated search documents *which* architectures were tried, making results more transparent than “we chose 3 layers with 64 units.”

Summary

Key takeaways:

1. Architecture choice **strongly** affects NN performance
2. Random search is a strong baseline — always try it first
3. Bayesian optimization is most sample-efficient
4. Hyperband / SHA offers the best speed/quality trade-off
5. Production: KerasTuner, Optuna, Ray Tune, Ax all wrap these

References:

- ▶ **Elsken, Metzen & Hutter (2019)** — *NAS: A Survey*, JMLR 20(55).
- ▶ Bergstra & Bengio (2012). *Random search*. JMLR 13.
- ▶ Li et al. (2018). *Hyperband*. JMLR 18(185).
- ▶ Garnett (2023). *Bayesian optimization*. CUP.
- ▶ Snoek et al. (2012). *Practical Bayesian optimization*. NeurIPS.